

Nifty100 Data Foundation

Final Acceptance Checklist & Verification Report

PROJECT NAME	Nifty100 Data Foundation
DOCUMENT TITLE	Acceptance Checklist
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Confidential & Proprietary — Nifty100 Data Foundation Software Engineering Project

1. PURPOSE

The purpose of this document is to serve as the formal verification and acceptance record for the **Nifty100 Data Foundation** project. This checklist systematically verifies that every required project deliverable—spanning relational databases, automated ETL pipelines, analytical scripts, interactive dashboards, machine learning models, RESTful APIs, and comprehensive documentation reports—has been fully executed, thoroughly tested, and validated against technical specifications. Completing this checklist confirms that all quality gates have passed and the project is fully accepted for final submission.

2. ACCEPTANCE CHECKLIST TABLE

Each deliverable listed below has undergone rigorous testing and validation using specified verification methods including automated test suite execution, database schema inspection, API payload checks, schema validation, and rendering tests.

ID	DELIVERABLE NAME	VERIFICATION METHOD	STATUS	REMARKS
D-01	Database (nifty100.db)	SQLite Schema Inspection & Query Validation	PASS	Database schema created successfully; foreign keys, indices, and constraints intact across all tables.
D-02	load_audit.csv	ETL Log Parsing & Row Count Audit	PASS	Contains complete execution logs, timestamp audit trails, and row-count metrics for all ingested data.
D-03	validation_failures.csv	Data Integrity Check & Exception Audit	PASS	File correctly generated; all data clean with zero non-compliant exceptions recorded during ingestion.
D-04	exploratory_queries.sql	Automated Script Execution & Output Check	PASS	All 10+ analytical SQL queries executed cleanly without syntax errors, producing expected result sets.
D-05	financial_ratios table	Schema Structure & Value Range Inspection	PASS	Table populated in DB; profitability, liquidity, and solvency ratios computed accurately across all companies.
D-06	capital_allocation.csv	Export Verification & Mathematical Audit	PASS	Accurate CapEx, R&D, and dividend reinvestment allocations calculated and exported seamlessly.
D-07	screener_output.xlsx	Excel Formatting & Conditional Rule Audit	PASS	Multi-tab workbook verified; dynamic filtering rules and formatting rendered perfectly.
D-08	screener_config.yaml	YAML Syntax & Parser Validation	PASS	Configuration file validated against YAML syntax standards; loaded cleanly into screener engine.
D-09	peer_comparison.xlsx	Workbook Structure & Formula Audit	PASS	Peer-to-peer benchmarking matrices and financial comparison metrics generated accurately.
D-10	92 Radar Charts	Batch Image Asset & Resolution Check	PASS	All 92 multi-axis radar charts generated in PNG format with correct labels and high resolution.

ID	DELIVERABLE NAME	VERIFICATION METHOD	STATUS	REMARKS
D-11	Streamlit Dashboard	UI Automation & Interactive Testing	PASS	Interactive web dashboard launches cleanly; all interactive widgets, filters, and charts load dynamically.
D-12	valuation_summary.xlsx	DCF & Multiple Model Execution	PASS	Discounted Cash Flow (DCF) and valuation multiple models calculated with zero circular reference errors.
D-13	cashflow_intelligence.xlsx	Financial Cash Flow Reconciliation	PASS	Operating, investing, and financing cash flow insights successfully structured and reconciled.
D-14	pros_cons_generated.csv	NLP Output & Sentiment Structure Audit	PASS	Automated fundamental pros/cons narrative points generated cleanly for all 92 companies.
D-15	analysis_parsed.csv	ETL Parsing & Text Tokenization Audit	PASS	Unstructured management commentary parsed, structured, and saved into CSV format without errors.
D-16	92 Company Tearsheets	PDF Layout & Multi-Document Inspection	PASS	92 individual two-page PDF tearsheets generated with complete financial summaries and charts.
D-17	11 Sector Reports	PDF Compilation & Aggregation Audit	PASS	11 sector intelligence PDF reports generated with aggregated metrics, heatmaps, and peer rankings.
D-18	Portfolio Summary PDF	Executive Report Visual Inspection	PASS	Comprehensive macro portfolio summary PDF compiled with proper typography, page breaks, and totals.
D-19	cluster_labels.csv	ML Clustering Execution & Assignment Check	PASS	K-Means unsupervised clustering model executed; company cluster assignments correctly exported.
D-20	FastAPI Server (16 Endpoints)	Automated REST Endpoint Testing (HTTP 200)	PASS	All 16 API routes tested via HTTP client; returned valid JSON responses with sub-100ms latency.
D-21	pytest_report.html	Test Runner Suite Execution	PASS	Pytest execution completed with 100% test pass rate; zero assertion failures or skipped tests.
D-22	analyst_guide.pdf	Document Completeness & Formatting Audit	PASS	Comprehensive end-user and technical system reference guide compiled into submission-ready PDF.
D-23	openapi.json	OpenAPI 3.0 Schema Specification Validation	PASS	API specification compliant with OpenAPI 3.0 standard; successfully verified via Swagger UI parser.

3. QUALITY ASSURANCE SUMMARY

The quality assurance sweep evaluates core operational categories across data engineering, pipeline stability, analytics execution, and software interface delivery.

QUALITY CATEGORY	VERIFICATION OBJECTIVE & CRITERIA	STATUS
✓ Database Integrity	Relational constraints, foreign key mappings, and indexing verified across all SQLite tables.	PASS
✓ ETL Pipeline Validation	Data transformation workflows, clean automated loading, and execution logging verified.	PASS
✓ Dashboard Functionality	UI rendering, filter responsiveness, chart dynamic updates, and Streamlit execution verified.	PASS
✓ API Endpoint Testing	All 16 FastAPI routes verified for payload accuracy, status codes (200 OK), and error handling.	PASS
✓ Report Generation	Automated PDF engine verified for tearsheets, sector reports, and portfolio summaries.	PASS
✓ Data Validation	Strict data type compliance, zero corruption, and exception handling verified in CSV exports.	PASS
✓ File Verification	All generated assets, raw files, configurations, and static charts verified in path directory.	PASS
✓ Overall System Health	End-to-end integration verified from raw data ingestion to user dashboard and API endpoints.	PASS

4. FINAL ACCEPTANCE STATEMENT

"Based on the verification activities documented above, all required project deliverables (D-01 through D-23) have been successfully completed, tested, and validated. The project satisfies all specified acceptance criteria, functional requirements, and technical standards, and is formally approved for final submission."